

AI Automation Project

Concise demonstration summary - generated from the supplied terminal output and latest HTML execution report

Purpose: turn a user story into Playwright tests with Gemini, run them through a local MCP server, and publish execution reports.

What the project demonstrates

The automation agent reads user story US-002 (Add Product to Cart), uses Gemini to create structured test cases, generates Pytest plus Playwright tests using the existing Page Objects, then executes generated tests through an MCP tool. The runner records pass/fail results and uses Gemini to explain failures.

Execution flow

1. Input	User story and prompt templates
2. AI generation	Gemini creates 3 test cases and 3 new story-specific tests
3. Automation	Pytest and Playwright execute against Sauce Demo
4. Results	MCP returns result JSON; JSON and HTML reports are saved

Key components

AutomationAgent orchestrates the pipeline. **GeminiClient** is the LLM boundary. **Page Objects** contain browser locators and actions. **TestRunner** invokes pytest, enriches failures with AI analysis, and writes reports. **MCP server** exposes `run_tests` and `read_test_report` as local tools.

Run result

Tests	Passed	Failed	Duration
6	4	2	28.80 seconds

The newly generated US-002 tests all passed (3 of 3). Two older generated tests also ran and failed, so the overall result is reported as failed.

Observed failures and interpretation

1. Non-specific locator: one legacy test asserts text on a locator that matches all six cart item names, causing Playwright strict-mode failure. **2. Intentional mismatch:** another legacy test adds one item but asserts a cart count of five. Gemini correctly identifies both causes and recommends a specific locator and a matching expected count.

Evidence provided with this document

Terminal output	Shows orchestration steps, generated test files, MCP response, pytest output, and Gemini failure analysis.
HTML report	Presents the 6-test summary, per-test status, and the detailed AI analyses for the 2 failures.